A cheaper $k = 4$ decomposition over $\mathbb{F}_{p^4}$: C_4 $2.5 \cdot 10^{10} \to 6.9 \cdot 10^7$, $S \div$ 400, crossover (model) $2^{135} \to 2^{103}$

2026-10-14. `research/gaudry_quartic_d4_cost_20261014/`. Protocol registered (commit `2be9ad3d7`) before any claim data; code frozen at `2b49f38d1`. **Class: engineering:** a constant in one decomposition. The exponent $q^{2-2/n}$ and the asymptote past the crossover are unchanged.

What changed

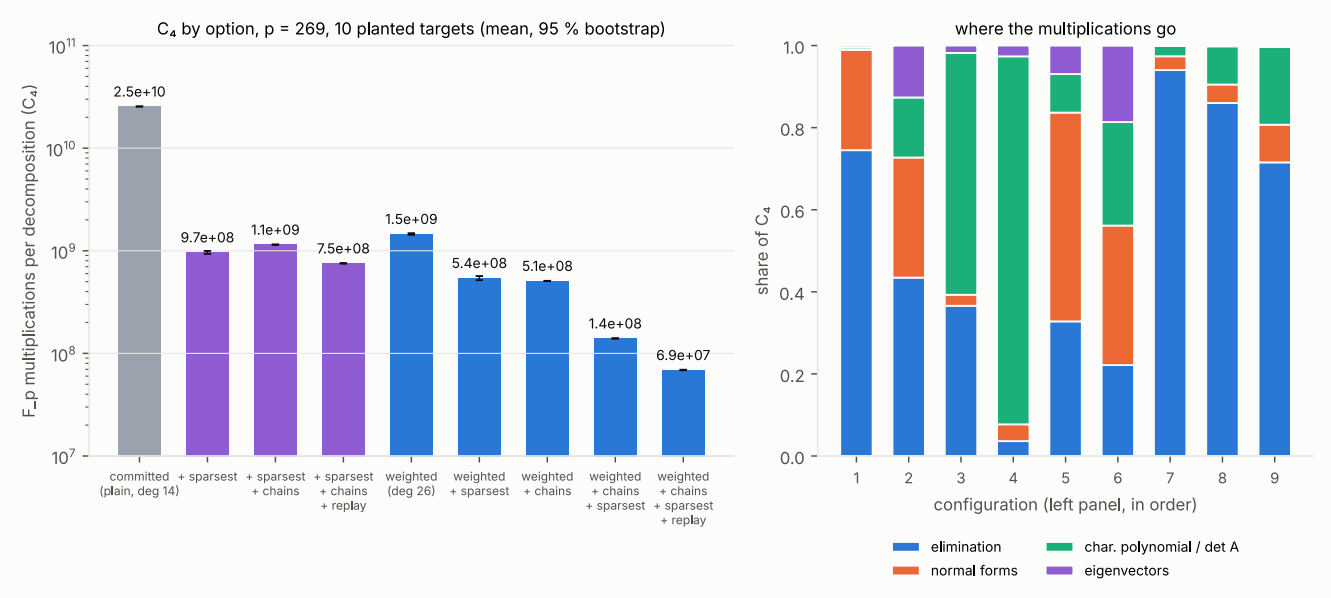
The `gaudry-d4` oracle (FGHR's D_4 system on a curve with rational 2-torsion) has four options. With all of them off it is the committed oracle, operation for operation (its committed counts were reproduced exactly).

- **weighted**: substitute $\sigma_4 = \pi^2$: four equations in $(\sigma_1, \sigma_2, \sigma_3, \pi)$ with weights $(2, 2, 2, 1)$, weighted degree 8. The Hilbert series is $(1 + t^2 + t^4 + t^6)^3 (1 + t + \ldots + t^7)$, so the weighted Macaulay degree is 26: a 3979×4200 matrix against 13412×11628.
- **chains**: only 44 of the 512 products πb leave the standard monomials. M_π is block-companion along π-chains, so the step needs 44 border normal forms, $\det A(\lambda)$ of a 44×44 polynomial matrix for the characteristic polynomial, and its kernels for the eigenvectors.
- **sparsest**: the sparsest candidate row is each elimination column's pivot.
- **replay** (F4Remake): rows that reduced to zero are skipped. A rank shortfall is topped up from the held rows up to the recorded generic rank, which a specialised system cannot exceed, so the result is exact.

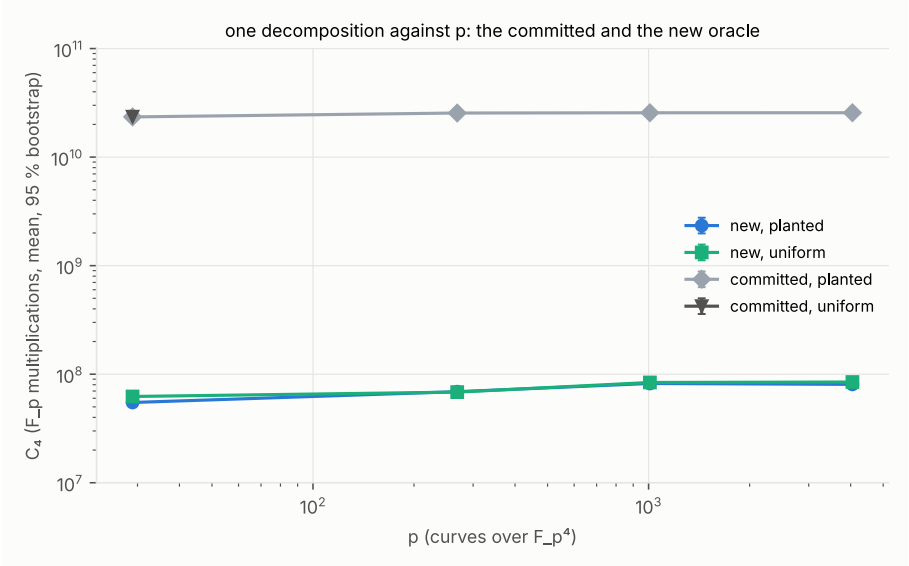
One decomposition (C_4, $\mathbb{F}_p$ multiplications; mean, 95 % bootstrap)

configuration ($p = 269$, 10 planted)	C_4	interval
committed (plain, degree 14)	$2.547 \cdot 10^{10}$	2.544–2.550
1. sparsest	$9.67 \cdot 10^8$	9.32–9.96
1. sparsest + chains + replay	$7.54 \cdot 10^8$	7.49–7.59
weighted (degree 26)	$1.459 \cdot 10^9$	1.433–1.483
weighted + chains + sparsest	$1.400 \cdot 10^8$	1.392–1.407
weighted + chains + sparsest + replay	$\mathbf{6.89 \cdot 10^7}$	6.85–6.93

Paired ratios: committed/new 369.8× (367.9–371.6). Sparsest pivots alone give 26.4×: most of the headline is pivot choice, which a competent eliminator on the old system would already have. The weighted form against the plain one at equal linear algebra gives **10.95**× (10.90–11.02). Across $p = 29, 269, 1009, 4049$ the new C_4 is 5.5–8.5 $\cdot 10^7$, 311–430× below the committed oracle on paired targets. All 648 closed solves equal the meet-in-the-middle enumeration, and paired targets give identical solution sets. 1 % of uniform targets at $p = 29$ do not close (weighted quotient of dimension 509–511). They are charged and dropped: re-solving one would cost about 370 new solves.



C_4 by option at $p = 269$ (left, log scale) and its split by phase (right). `raw/summary.json`.

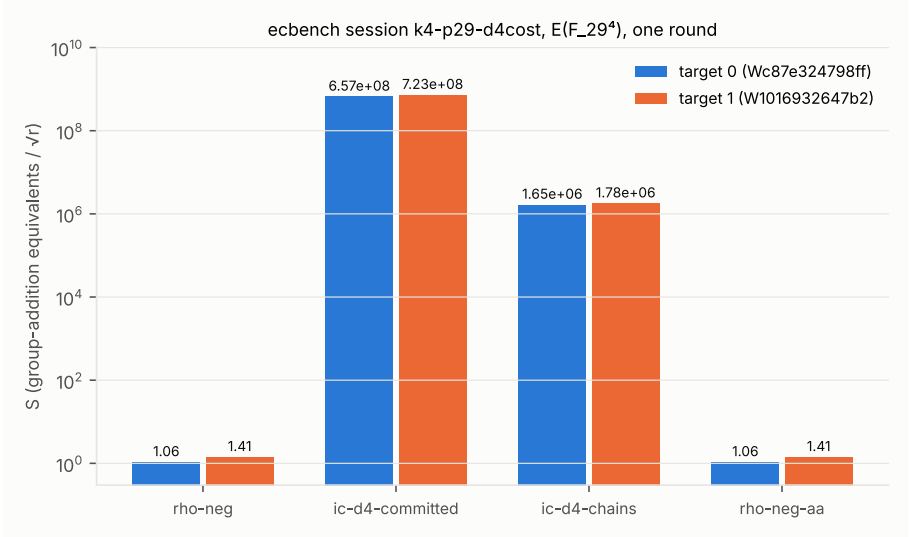


C_4 against p; planted and uniform targets; the committed oracle on the same targets.

End to end (ecbench session `ECBS1h4687f1d786c2`, $E(\mathbb{F}_{29^4})$, $\#E = 16 \cdot 44263$)

arm	target 0 S	target 1 S	trials	unsolved (charged)
rho, negation map	1.065	1.412		
IC, committed oracle	$6.571 \cdot 10^8$	$7.228 \cdot 10^8$	574 / 635	9 / 8
IC, new oracle	$\mathbf{1.653 \cdot 10^6}$	$\mathbf{1.780 \cdot 10^6}$	574 / 635	11 / 13

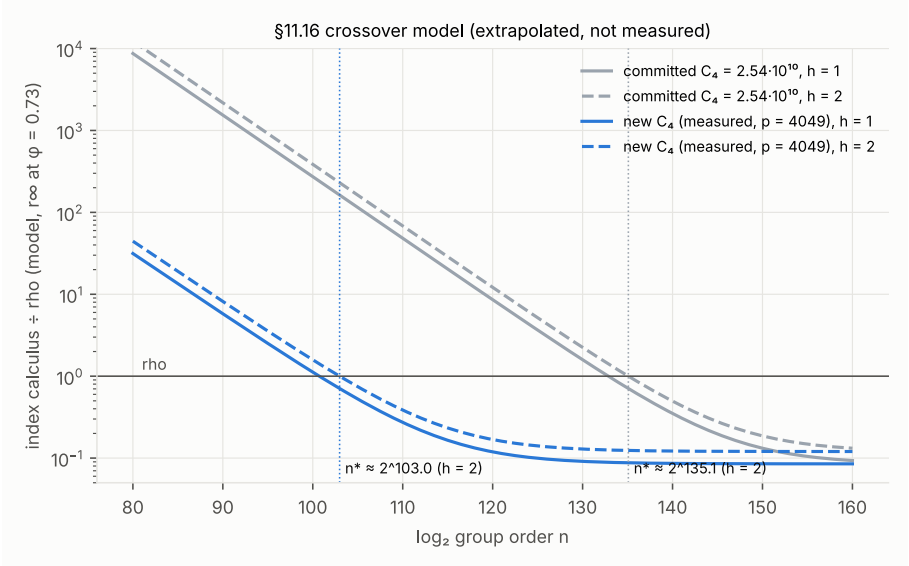
New/committed: **398× and 406×**, with identical trials and relations. IC still costs $1.39 \cdot 10^6$× rho (1.26–1.55) at this size. All 8 runs verified. The IC runs graded L1, so wall time (221 s against 7,321 s) is descriptive.



S per arm and target, session `k4-p29-d4cost`.

Crossover (§11.16's model, extrapolated)

$S/\text{rho} = \sqrt{h}(181(p/4)C_4/(126p^2) + r_\infty/4)$, with C_4 measured at $p = 4049$ and root finding extrapolated in $\log p$. At $h = 2$ the crossover moves from $2^{135.1}$–$2^{135.8}$ to $\mathbf{2^{103.0}}$–$\mathbf{2^{103.7}}$ (φ = 0.73 … 1); at $h = 1$, $2^{100.7}$–$2^{101.2}$. Past it the ratio still tends to 0.12–0.22× rho. This is a model, not a measured S.



The model with the committed and the new C_4.

Predictions and limits

All registered predictions held: A1 exact; A2 C_4 in range and ≥ 200× below; A3 ≤ 3 % unsolved; A4 26× and 11×; B2 identical trials; B3 398×, 406×; C1 2^{103}. B1: verified, with the replay audit in `audits/`.

What did not work: weighted degrees below 26; chains on the plain form, where σ_1 has 260 chains and det A costs more than the dense characteristic polynomial; replay without top-up, which fell back on 42 % of systems at $p = 29$; and solving the degenerate systems.

Not tried: F4 with weighted sugar, Wiedemann, Magma on the weighted system.

No `docs/bounds` domain exists for $\mathbb{F}_{p^4}$ curves, so this is a paired session, not a challenge verdict.

Environment: Claude Code cloud container, Xeon @ 2.10 GHz, 4 vCPU, rustc 1.97.0. Conductor task T-1.